CycleMate

Helping IVF patients finish the treatment they started

A web-based companion for the IVF cycle · patient app + coordinator view · EN / UZ / RU

AI in Medicine & Biostatistics
Mid-term project · My (Digital) Health App
MS4 - 260721 Weonchul Shin · October 2026

1 · THE HEALTH PROBLEM

Infertility affects about **1 in 6 adults** worldwide (WHO, 2023). IVF is effective — but only when a patient completes the cycle.

The gap: up to **1 in 5 patients drop out** of IVF before finishing, and the leading reason is not medical failure. It is the **burden of treatment itself**. Most who stop never return.

17%	19–28%	18–27%
of couples drop out mid-treatment	**cite physical / psychological burden as the reason**	**show clinical anxiety or depression during IVF**
Verberg 2008, n=384	Gameiro 2012 (n=21,453); Verberg 2008	China 2025, n=2,465

The IVF journey, in one line. Baseline scan → 8–12 days of stimulation injections, monitored by scans and blood tests → trigger shot → egg retrieval → fertilisation in the lab → embryo transfer → two-week wait → pregnancy test. About 28 days; most of it spent at home between visits.

Why a leaflet is not enough. IVF is not one protocol. During stimulation the dose and timing of each injection are adjusted to the patient's own hormone response, scan by scan — the plan for day 9 is only known on day 8. Every patient therefore walks a different, moving schedule, and no clinic can re-explain it by phone to every patient every day. The gap between "what the doctor decided" and "what I'm supposed to do tomorrow" is where confusion, anxiety and drop-out begin.

WHAT "BURDEN" ACTUALLY IS

From clinic experience in Tashkent and the literature, four things wear patients down:

- Uncertainty — "what happens next?"
- Daily load — visits, pills, timing
- Feeling alone between visits
- Nowhere to ask a small question

TARGET USERS

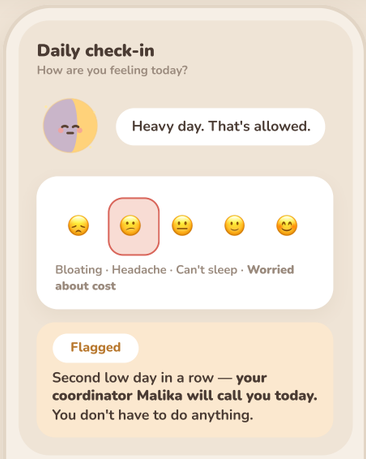
- **Patients** going through an IVF cycle — primary user. Many live far from Tashkent and cannot visit daily.
- **Clinic coordinators** — see who needs a call first, instead of waiting for patients to reach out.
- **Doctors** — receive only the questions that need a medical answer.

2 · THE APP — ONE BURDEN, ONE FEATURE

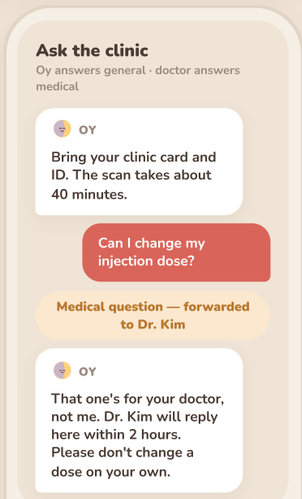
Uncertainty "what happens next?"	**Cycle timeline** — today's day, next step, estimated dates; shown as a moon that waxes with the cycle
Daily load	**Today's plan** — never more than 3 tasks; clinic visits, home medication, trigger-shot countdown
Feeling alone	**10-second check-in** — mood 1–5; two low days in a row and the coordinator calls *first*
Nowhere to ask	**Ask the clinic** — Oy answers general questions; anything medical is forwarded to the doctor
Cost worry	**Running total** — paid so far and estimated remaining; no surprises



Home
where am I in the cycle?



Check-in
the clinic reaches out first



Ask
AI knows what it must not answer

HOW IT WORKS

Patient
web app · any phone
phone-number login

⇄

Clinic server
encrypted · on-premise
consent-scoped sharing

⇄

Coordinator & doctor
call list · doctor queue

AI 1 · QUESTION ROUTER
Classifies each message as *general* (Oy answers from the clinic FAQ) or *medical* (forwarded to the doctor). Unknown → forwarded, never guessed.

AI 2 · CALL-LIST RANKING
Orders the coordinator's list from mood trend, missed tasks and unanswered questions. It changes who gets called first — never the treatment.

The AI never diagnoses, never changes a dose, never interprets a result. Every clinical decision stays with the medical team.

MEET OY — WHY A MOON



A lunar month and an IVF cycle are both about 28 days. Oy waxes through stimulation, is full at embryo transfer, and wanes toward the pregnancy test. One glance at the moon tells a patient where she is — no calendar, no reading.



Calm by design. Medical apps often use alerts, red badges and countdowns that raise anxiety. Oy replaces them with a soft, familiar companion who reacts to how the patient feels — so the app lowers fear instead of adding to it.

A moon, not an embryo. Oy stands for the *cycle*, not the outcome. Patients are never asked to bond with an embryo; if a cycle fails, the moon simply begins again.

A word for today. Below the task list sits one of 28 short lines, one per cycle day — e.g. day 7: *"You are doing better than you think you are."*, day 19: *"You've come a long way. Let yourself take a slow breath."* They never promise an outcome; they only keep the patient company.



reacts to check-ins, tasks and questions

3 · PRIVACY, SECURITY & ETHICS

PRIVACY & SECURITY
Login only with the phone number registered at the clinic (SMS code). Data encrypted in transit and at rest, stored on the clinic's own server. Patient chooses what to share — partner sees schedule only, never mood or messages. Deleted 12 months after the cycle, or on request.

ETHICS
AI gives general information only; medical questions go to a human. When unsure, it forwards rather than guesses. Flags prioritise a coordinator's calls — they never label or diagnose a patient. The companion is designed to reduce fear, not to build attachment to an outcome.

EQUITY & ACCESS

- **Web-based, no app store** — runs on any low-end phone; reminders fall back to SMS when offline.
- **Built for patients outside Tashkent** — only 4 visits are essential; remote check-ins and partner clinics cover the rest.
- **EN / UZ / RU** — icons and one task per screen for low-literacy users.

FEEDBACK → CHANGE · IMPACT · NEXT

Clinic coordinator: "Patients who are struggling rarely message us — they just stop coming." → **Added the 2-low-days rule:** the coordinator is flagged and calls first, instead of waiting for the patient.

v1 → v3 (self-review): a plain timer-style UI felt clinical and cold → added Oy, warm palette, one task at a time.

PRIMARY	SECONDARY	CLINIC
cycle completion rate	**check-in trend · time-to-contact**	**proactive calls made**